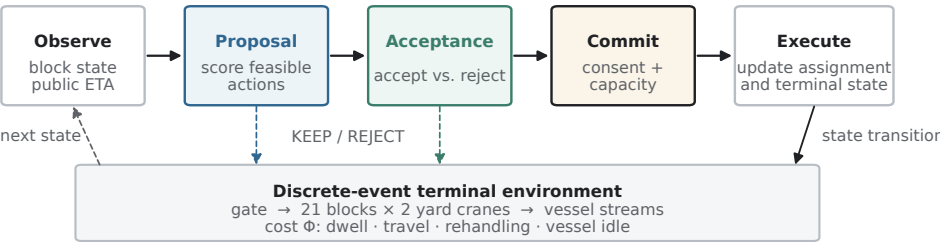
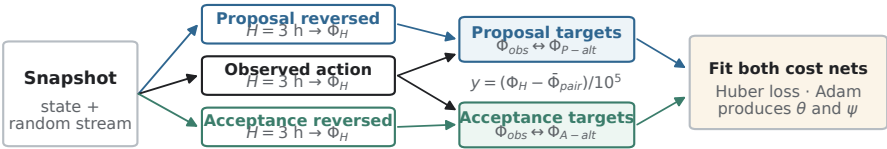


(a) Online inference — every 60 s



(b) Offline learning — counterfactual simulation only



Counterfactual branches generate labels offline; online decisions use only the fitted networks.